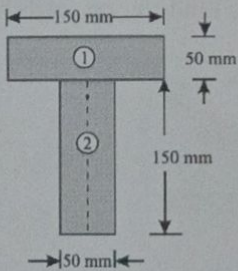
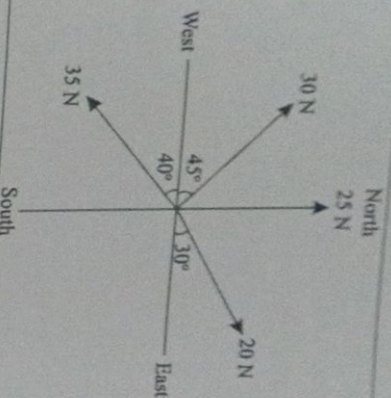


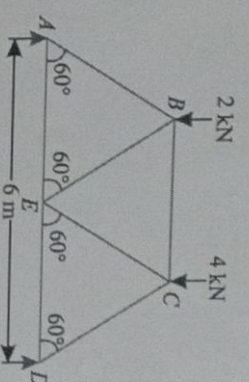
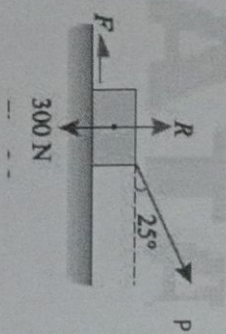
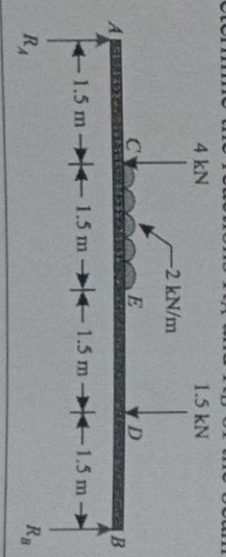
	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of BSHM Summer term Examination Academic Year: 2024-25		
Class: FY	Subject: Engineering Mechanics	Subject Code: ESC 109	
Date: 23.6.2025	Time: 10:00 to 13:00	Max. Marks: 100	

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- All questions are compulsory

Q.N.		Marks	CO	BT
Q.1	a Explain the following i) Assumptions in analysing trusses ii) Laws of friction	10	CO4	BT1
Q.2	a Explain the following i) Perpendicular axis theorem ii) Types of loads iii) Equation of projectile iv) Velocity v) Angle of friction	10	CO1	BT2
Q.3	a Evaluate the moment of inertia of a T-section with flange as 150 mm × 50 mm and web as 150 mm × 50 mm about X-X and Y-Y axes through the centroid of the section. 	10	CO3	BT5
Q.4	a The following forces act at a point: Find the magnitude and direction of the resultant force.	10	CO1	BT4



Q.5	a	A particle, starting from rest, moves in a straight line, whose equation of motion is given by: $s = t^3 - 2t^2 + 3$. Find the velocity and acceleration of the particle after 5 seconds.	10	CO5	BT4
Q.6	a	A ball is dropped from the top of a building 40 m high. At the same time, another ball is thrown upwards from the foot of the building with a velocity of 20 m/s. When and where the two stones cross each other?	10	CO5	BT4
Q.7	a	A motor car takes 15 seconds to cover 30 meters and 20 seconds to cover 42 meters. Find the uniform acceleration of the car and its velocity at the end of 18 seconds. Calculate forces in the members of truss.	10	CO6	BT4
Q.8	a		10	CO2	BT5
Q.9	a	A body of weight 300 N is lying on a rough horizontal plane having a coefficient of friction as 0.3. Find the magnitude of the force P. 	10	CO2	BT4
Q.10	a	A simply supported beam, AB of span 6 m is loaded as shown in Figure. Determine the reactions R_A and R_B of the beam. 	10	CO1	BT4



SGGS Institute of Engineering and Technology, Nanded

Summer Term Examination 2024-25

F.Y. B. Tech.

All Branches

Programming for Problem

ESC 103/107

Solving

Date: 20/06/25

Time : 10:00am -1:00pm

Max. Marks : 100 Marks

- Note:
- Attempt **any 10** questions from the following.
 - Assume suitable data if necessary
 - Use of non-programmable calculator is not permitted.
 - This paper contains 2 pages.

Q.N.	Questions	Marks	CO	BT
Q.1	Draw the flowchart and Write a C program to find the largest of 3 numbers.	10	CO1	BT3
Q.2	Explain any five operators used in C language.	10	CO2	BT2
Q.3	Write a calculator program in C language to do simple operations like addition, subtraction, multiplication and division. Use switch statement in your program.	10	CO2	BT3
Q.4	Write a C program to check whether a given number is palindrome or not.	10	CO4	BT3
Q.5	Differentiate the following: 1) call by value and call by reference 2) Exit controlled loop and Entry controlled loop.	10	CO4	BT4
Q.6	What is array? Write a C program for Bubble sort.	10	CO4	BT4
Q.7	Write a C program to read N numbers into an array & perform Linear search.	10	CO5	BT4
Q.8	Explain recursion in C. Write a C program in C to find Factorial of a given number using Recursion.	10	CO4	BT4
Q.9	Define string. Explain the following string functions in c with suitable program: 1) strcmp() 2) strcat()	10	CO5	BT4
Q.10	Write a program to maintain a record of a student with entities as student_name, student_no, student_marks, student_div.	10	CO5	BT3
Q.11	Explain different modes in which we can access a file using suitable programs.	10	CO5	BT3



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Production Engineering
Summer Term Examination

Class: First Year (Gen.)

Academic Year: 2024-25

Date: 21/06/2025

Subject: Engineering Graphics & Design

Subject Code: ESC102/106

Time: 10:00 - 13:00

Max. Marks: 100

Note

- Attempt all questions.
- Figures to right indicate full marks.

Q. 1	Solve the following questions.	M	CO	BT
a)	Draw a regular Hexagon of 60 mm side, using General method.	05	CO1	BT1
b)	Construct a regular pentagon of 50 mm diameter.	05	CO1	
c)	Draw the projection of point A situated at 30 mm away from HP and 45 mm from VP when it exists in (A) first quadrant (B) Second quadrant (C) Third quadrant (D) Forth quadrant.	05	CO1	
d)	A point R is 30 mm above the HP and 45 mm in front of VP draw it's all possible views (LHSV & RHSV).	05	CO1	
e)	What are the advantages and disadvantages of using CAD systems to create engineering drawings?	05	CO4	
f)	Explain the various basic editing commands used in Auto CAD. (Any Four).	05	CO4	
Q. 2	Solve the following questions. (Any One A or B)			
A	Draw the projections of a line AB 100 mm long inclined at 45° to VP and 30° to HP. One end of the line is 20 mm above HP and in VP. Determine apparent lengths and inclinations. Also mark the traces.	12	CO1	BT3
B	The front view and top view of a straight-line PQ measures 50 mm and 65 mm, respectively. The point P is on HP and 20 mm in front of the VP. The front view of the line is inclined at 45° to the reference line. Determine the true length of PQ and its true inclination with the reference planes. Also locate the trace.	12		BT3
Q. 3	Solve the following questions. (Any One A or B)			
A	A square lamina of 40 mm side rests on one of its sides on HP. The lamina makes 30° to HP and the side on which it rests makes 45° to VP. Draw its projections.	12	CO1	BT3
B	An equilateral triangular lamina of 50 mm sides lies with one of its edges on HP such that the surface of the lamina is inclined to HP at 60° The edge on which it rests is inclined to VP at 45° Draw its projections	12		BT3
Q. 4	Solve the following questions. (Any One A or B)			
A	A square prism 35mm sides of base and 65 mm axis length rests on HP on one of its edges of the base which is inclined to VP at 30° . Draw the projections of the prism when the axis is inclined to HP at 45° .	14	CO1	BT3
B	A cylinder 35 mm diameter and 60 mm axis, its resting on one point of a base circle on HP while its axis makes 45° with HP and T.V. of the axis 35° with VP. Draw projections.	14		



Class: First Year (Gen.)

Subject: Engineering Graphics & Design

Subject Code: ESC102/106

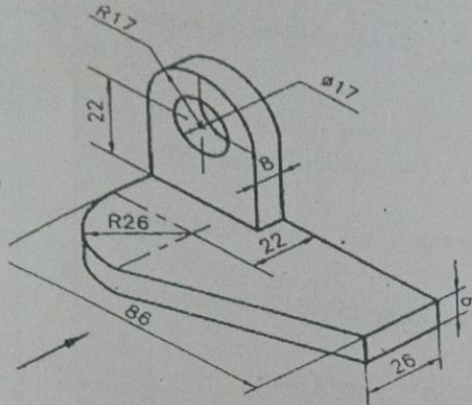
Date: 21/06/2025

Time: 10:00 - 13:00

Max. Marks: 100

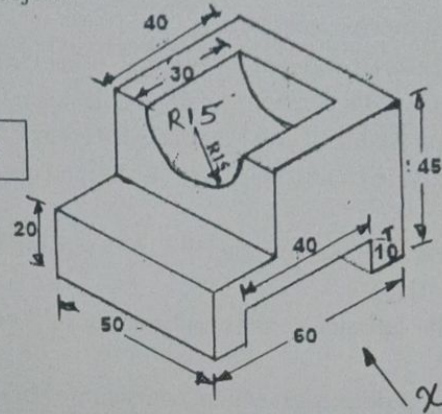
Q.5

a) Using First Angle Projection Method draw – (a) Front View (b) Top View (c) Right Hand Side View for the following object.



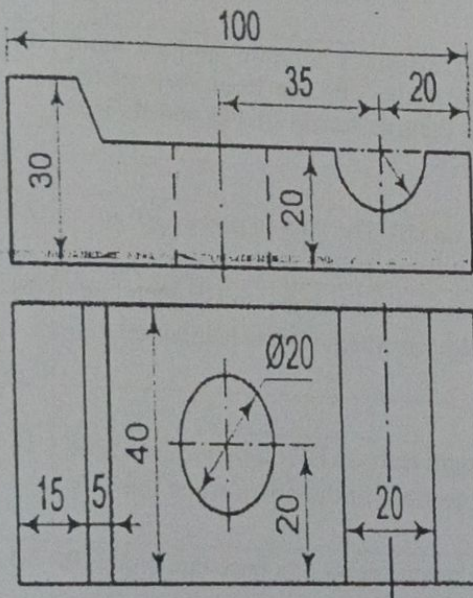
OR

b) Using First Angle Projection Method draw – (a) Front View (b) Top View (c) Left Hand Side View for the following object.



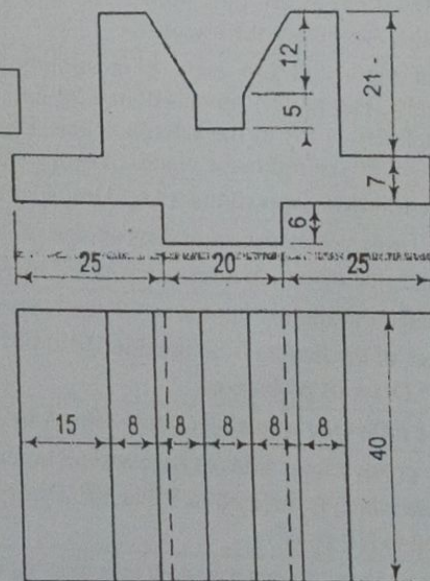
Q.6

a) The Front and Top view of an object are shown in fig. Draw its isometric view.



OR

b) The Front and Top view of an object are shown in fig. Draw its isometric view.



Course Outcomes:

CO1. Draw projections of point, line, plane and solid using the fundamental principles.

CO2. Read, interpret, and draw orthographic projections.

CO3. Read, interpret, and draw isometric projections.

CO4. Draft various Geometrical Elements used in Engineering Practice using CAD software.

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

Summer Term Examination 2024-25

F.Y. B. Tech

Div: All (A to J)

Department: BSH&M

Subject: Chemistry

BSC102

Max. Marks: 100

Date: 19/06/2025

Time: 10.00 am - 1.00 pm

Note: i. Attempt all questions

iii. Use of a non-programmable calculator is permitted.

ii. Assume suitable data if necessary.

iv. This paper contains 2 pages.

Data: Atomic Weight: Ca = 40, H = 1, O = 16, Cl = 35.5, S = 32, Mg = 24, N = 14, Si = 28, C = 12

Q.N.	1	Questions	Marks	CO	BT
		Question 1 is Compulsory	15		
			3	CO4	BT4
a		A sample of coal contains C = 82 %, O = 8 %, H = 6 %, S = 0.5, N = 0.2 %.. Calculate GCV & NCV of a coal sample using Dulong & Petit formula.	3	CO2	BT2
b		Classify the following impurities in to temporary, permanent and non-hardness causing impurities $\text{Ca}(\text{HCO}_3)_2$, MgSO_4 , CaCl_2 , NaCl , $\text{Mg}(\text{HCO}_3)_2$, CaSO_4 .	3	CO3	BT1
c		Enlist any six principles of Green Chemistry.	3	CO5	BT2
d		Differentiate between Galvanising and Tinning.	3	CO1	BT1
e		Enlist the important applications of carbon nanotubes in various fields.	3		
		Attempt any three	12		
Q.N	2		4	CO2	BT3
a		Define corrosion. Explain how the rate of corrosion of the following metal is influenced by atmospheric oxygen. i) Molybdenum ii) Sodium	4	CO4	BT4
b		A KMnO_4 solution of unknown concentration is placed into a spectrophotometer. A student finds that the solution's absorbance is 0.920. The molar absorptivity of KMnO_4 is $7.35 \text{ Lit mol}^{-1} \text{ cm}^{-1}$ and the path length of the solution container is 5.00 cm. What is the concentration of solution?	4	CO3	BT2
c		Give the important properties & uses of Fullerene	4	CO1	BT1
d		Find the acid value of an oil sample whose 6 ml required 0.02 N KOH for titration. (Density of the Oil = 0.91). State whether the oil is suitable for lubrication or not?	4		
		Attempt any three	15		
Q.N.	3		5	CO4	BT4
a		Define saponification value, give the reaction for saponification. Find the saponification value of a vegetable oil sample, whose 5 gms was saponified using excess alcoholic KOH (0.5 N). The mixture required 15 ml of 0.5 N HCl while blank titration required 45.0 ml of same HCl.	5	CO3	BT2
b		A solution of KMnO_4 of concentration $3.5 \times 10^{-5} \text{ M}$ shows an absorbance of 1.3 at its wavelength of λ_{max} . Calculate the concentration of an unknown KMnO_4 solution which shows an absorbance of 0.86 in the same cuvette.	5	CO2	BT1
c		Distinguish between vulcanized and non-vulcanized rubber.	5	CO1	BT4
d		Explain with a neat diagram the zeolite process of water softening including the following points. i) Principle, ii) Process, iii) softening & regeneration reactions.	5		
		Attempt any three	18		
Q.N	4		6	CO5	BT3
a		Define Lubricant & Lubrication. Explain thick film fluid mechanism of lubrication with neat labelled diagrams.	6		

b	1.50 gm of coal sample was taken for C and H estimation by combustion method. The increase in weight of tube containing anhydrous CaCl_2 and KOH bulb was found to be 1.25 gm and 3.92 gm respectively. Calculate the % of C and H.	6	CO3	BT3
c	State Pilling Bedworth rule, Explain different types of oxide layer formed during dry corrosion.	6	CO2	BT2
d	1.2 gm of CaCO_3 was dissolved in HCl and the solution was made to 1000 ml with distilled water. 20 mL of the above solution required 35 mL of EDTA solution on titration. 50 mL of the hard water sample required 30 mL of the same EDTA solution on titration. After boiling 50 mL of this water, required 12.5 mL of EDTA solution, Calculate temporary, permanent, and total hardness.	6	CO4	BT4
Q.N.	5	Attempt any Four	40	
a	What is proximate analysis? Give its Significance. 2.5 g of coal sample was exactly weighed into the silica crucible. After heating for 1 hour at 110°C the residue weighed 2.368 g, the crucible next covered with a vented lid and strongly heated for exactly seven minutes at $950 \pm 20^\circ\text{C}$. The residue weighed 1.75g. The crucible was then heated without cover at 700°C until a constant weight was obtained. The last residue was found to weigh 0.95g. Calculate the percentage results of above analysis. (% Moisture, % Volatile matter, % Ash and % F.C.).	10	CO2	BT2
b	i) 2.5 gm of blended oil was saponified using excess alcoholic KOH (0.5 N). After refluxing the mixture was titrated against 0.5 N HCl. The burette reading was found to be 24 ml. The blank titration required 40 ml of the same HCl solution. Find the saponification value of the oil. If the oil used for blending has a saponification value of 188, calculate % oil in the blend. ii) 2.5 gm of coal sample was analyzed for nitrogen content by Kjeldhal's method. The liberated ammonia required 12.7 ml of 0.5 N H_2SO_4 solution for neutralization. In a separate experiment, using a bomb calorimeter 1.5 gm of coal sample gave 0.28 gm of BaSO_4 . Calculate % N & S in the coal sample.	10	CO3	BT3
c	What are the different methods for controlling corrosion? Explain cathodic protection method in detail along with its type.	10	CO5	BT2
d	Give the preparation, properties and uses of following polymers 1) PMMA 2) Kevlar	10	CO3	BT1
e	What are the advantages & Disadvantages of Lime & Soda method? Calculate the quantity of lime (90 % Pure) and soda (95% pure) required for softening of 50000 liters of water containing the following impurities. $\text{Ca}(\text{HCO}_3)_2 = 81$ ppm, $\text{MgCl}_2 = 95$ ppm, $\text{CaSO}_4 = 68$ ppm, $\text{SiO}_2 = 50$ ppm, $\text{Mg}(\text{HCO}_3)_2 = 146$ ppm, $\text{H}_2\text{SO}_4 = 49$ ppm.	10	CO1	BT4

Course Outcomes (CO)

- CO1. Students will be able to understand various industrial processes and products.
CO2. Enabling students to know about the relevant engineering materials, their chemistry and recent technology.
CO3. Outer world will be benefited with current scientific knowledge.
CO4. Students will be able to understand various analytical methods used for material analysis.
CO5. To strengthen the student's understanding to the related core subjects in their respective discipline.

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of BSH&M Summer Term Examination Academic Year: 2024-25	
Class: FY B.Tech.	Subject: Professional Communication	Subject Code: AES102/HMC110
Date: 25.06.2025	Time: 10:00am to 01:00pm	Max. Marks: 100

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange Sharing of stationery, calculator etc. not allowed.
- iv. Figures to the right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.	Attempt all of the following questions.			Marks	CO	BT
Q.1	A	Do as directed. (Any five)	10	CO1	BT1	
	1	I have to choose ... the two prizes. (between/among/amongst)				
	2	I would travel by taxi. (Write the sentence structure)				
	3	He bought ... pen ... and ink ... for ... union. (Use articles)				
	4	When will he learn these things? (Change into passive voice)				
	5	I said, "Let's go to cinema." (Change into indirect narration)				
	6	Scientists have discovered medicine for COVID. (Change into simple present tense.)				
	B	Correct the following sentences by rewriting it. (Any five)	10	CO1	BT1	
	1	I am having four shirts and three pants.				
	2	Ask her to sit besides me.				
	3	Did you went to the park?				
	4	He could not decide among the two choices.				
	5	My younger brother is good in mathematics.				
	6	Monkeys are climbing about tree.				
Q.2	A	United India Ltd. is looking for young, dynamic and result oriented engineers from all branches. Write an application with CV to the CEO, United India Ltd., Bangalore.	10	CO2	BT2	
	B	Develop a well-knit paragraph on any one of the topics: 'Beat plastic pollution' OR 'Give blood, give hope'	10	CO2	BT3	
Q.3		Write an order for 60 computers from Microbe Technologies, Chennai by providing all details of the quotation you received.	10	CO3	BT3	
Q.4		As a CR of the class, write a report focusing the causes related to the high absenteeism of the students to the class.	10	CO3	BT4	
Q.6	A	Assume that you are the head of the Department of Mechanical Engineering. You need some new and updated equipment for your mechanical lab. Write a letter of enquiry to Moto-Pro Ltd., Indore giving full details of your requirements in complete block style.	10	CO4	BT4	
	B	Suman Textiles Co. Ltd. Hyderabad has been facing a decline in profits by various reasons as fall in sales. As a marketing manager, you have been asked by the Managing Director to investigate into causes and make suitable recommendations as a report.	10	CO3	BT6	
Q.7		Write a precis of the following paragraph with suitable title.	10	CO4	BT5	

India may be making rapid progress in IT industry and in primary education, but with the absence of value-based education they seem to have neither the motivation nor the wisdom to move in a joyous direction. These values are not readily available through the television either. Religion, which should have been the source of imbibing values has become dangerously ritualistic. Moreover, all religions actively advance irrelevant logic against other religions. Books can provide joyous wisdom. It is surprising that no one seems to be actively seeking it through the printed media. Almost all Indians have lost the pleasure of reading the languages. In fact, they do not even seem to be aware that there is an endless treasure waiting to be tapped through the books in Indian languages. It is up to the intellectuals to provide leadership and inspiration to those who cannot master the necessary motivation to read books in Indian languages

Q.8

As a Lab in charge write a letter of complaint to M/S Das & Co. for the delivery of defective equipment for your laboratory costs about 2 Lakh asking them about replacement.

10 CO4 BT6

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

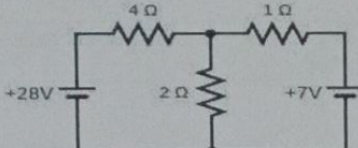
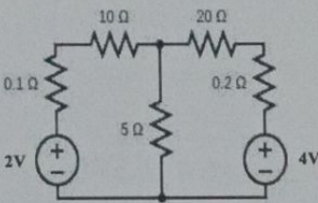
Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	To enable students to learn the structure and style of effective sentences.
CO 2	To enable students to express themselves in English effectively.
CO 3	To enhance students' Listening, Speaking, Reading & Writing Skills.
CO 4	To help students to understand nuances of technical writing.

		
Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Electrical Engineering Summer Term Examination Academic Year: 2024-25		
Class: First Year	Subject: Basic Electrical Engineering	Subject Code: ESC 101/105
Date: 24/06/2025	Time: 10:00 to 13:00	Max. Marks: 100

Note:

- All questions are compulsory
- Step-wise marks are given for numerical questions.
- A scientific calculator is allowed.
- This paper contains 02 pages(back to back)

Q.N.	All questions are compulsory		Marks	CO	BT
Q1	a)	i) State and explain ohms law and what are its limitations ii) State and explain Kirchhoff's both laws.	6	CO-1	1
	b)	What are the types of electrical sources? Explain Independent and Dependent sources with symbols.	6	CO-1	1
	c)	Define i) Current, ii) Voltage , iii) Electrical Power, iv) Resistance	4	CO-1	1
Q.2	a)	State Superposition theorem and write steps to solve the Superposition theorem	7	Co-2	2
	b)	Write steps to solve mesh Analysis and Node analysis with a circuit diagram	7	CO-2	2
	c)	Find the current through 2Ω resistor using Thevenin's theorem. 	7	CO-2	3
	d)	Find the current through 5Ω resistor using Norton's theorem. 	7	CO-2	3
Q.3	a)	Compare Electric circuits and Magnetic circuits	8	CO-3	1
	b)	State Faraday's laws of electromagnetic induction. (Both the Laws	4	CO-3	1
	c)	State Fleming's left and Right-hand Rule	4	CO-3	1
	d)	An iron ring with a mean circumference of 0.8 meters is uniformly wound with 550 wire turns. Calculate the flux density value that a current of 1.1 Amp would produce in the ring. (Assume $\mu_r = 1400$)	5	CO-3	3
Q.4	a)	Explain the advantages of a phase system over a single-phase system.	6	CO-4	2

	b)	Define i) Cycle ii) RMS Value iii) Instantaneous Value iv) Frequency	4	CO-4	
	c)	A Resistance of $5\ \Omega$ and Capacitive reactance of $10\ \Omega$ are connected in series if the current through the circuit is 3A; find impedance, phase angle, and power factor.	5	CO-4	3
Q.5	a)	Draw and explain the construction of single-phase transformers and their application	6	CO-5	2
	b)	Explain Open circuit (OC) and Short circuit (SC) test of 1 phase transformer	7	CO-5	2
	c)	Draw and Explain the working principle of a Permanent Magnet Moving Coil (PMMC) meter with a neat diagram.	7	CO-5	2

————— ALL THE BEST —————

Bloom's Taxonomy (BT):-

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:	
CO 1	Describe the basic concepts of voltage, current, EMF, energy sources, and fundamental laws like Ohm's and Kirchhoff's in electric circuits.	
CO 2	Analyze DC electrical circuits using series-parallel combinations, mesh and nodal analysis, and theorems such as Superposition, Thevenin's, and Norton's.	
CO 3	Apply electromagnetic laws to analyse magnetic circuits, induced EMF, and energy storage in magnetic fields	
CO 4	Analyze single-phase and three-phase AC circuits using phasor diagrams, and calculate impedance, power factor, and power components.	
CO 5	Explain the construction and operation of single-phase transformers and basic electrical measuring instruments like PMMC meters and wattmeters.	

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of BSH&M (Mathematics) Summer term Examination, Academic Year: 2024-25 (Semester – II)		
	Class: FY B.Tech.	Subject: Mathematics-II (Multivariable Calculus, Partial Differential Equations, Laplace Transforms)	Subject Code: BSC104
	Date: 17/06/2025	Time: 10.00-13.00	Max. Marks: 100
	Note: i) calculators are not allowed. ii) This paper contains total 2 pages.		

Q.1	Attempt any five	marks	CO	BT
a	Find the partial derivatives $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$ & $\frac{\partial^2 f}{\partial x \partial y}$ for the function $f(x, y) = \ln(x^2 + y^2) + \tan^{-1}\left(\frac{y}{x}\right)$.	06	CO1	BT1
b	Express $\frac{\partial z}{\partial u}$ & $\frac{\partial z}{\partial v}$ as a function of u & v by using the Chain Rule, if $Z = \tan^{-1}\left(\frac{x}{y}\right)$, $x = u \cos(v)$, $y = u \sin(v)$.	06	CO1	BT2
c	If $f(x, y) = \frac{x-y}{xy+2}$ then find i) The directions in which $f(x, y)$ increases and decreases most rapidly at $p_0(1, -1)$. ii) The directional derivative of $f(x, y)$ in the direction of $\vec{u} = 2i - j$ at $p_0(1, -1)$.	06	CO1	BT2
d	Find equation of tangent plane and normal line to the surface $x \ln(y) + 2y \ln(z) - x = 0$ at the point $p_0(2, 1, e)$.	06	CO1	BT2
e	Find all the local maxima, local minima & saddle points (if any) of the function $f(x, y) = 6x^2 - 2x^3 + 3y^2 + 6xy$.	06	CO1	BT3
f	Find solution of following IVP by using Laplace transform. $y'' + y' - 6y = 0$, $y(0) = 0$, $y'(0) = 1$.	06	CO4	BT3
Q.2	Attempt any five			
a	Evaluate the integral $\int_0^1 \int_0^1 (xy)^2 \sin(\pi x^3) dx dy$.	06	CO1	BT2
b	Sketch the region of integration, reverse the order of integration and evaluate the integral $\int_0^\pi \int_x^\pi \frac{\sin y}{y} dy dx$.	06	CO1	BT2
c	Evaluate the following integration over the given box B. $\iiint_B z e^{x+y} dx dy dz$, $B = [0, 1] \times [0, 1] \times [0, 1]$.	06	CO1	BT2

d	Let curve C is $c(t) = (\sin t, \cos t, t)$, with $0 \leq t \leq \pi$ and $F(x, y, z) = xi + yj + zk$ compute $\int_C F \cdot ds$.	06	CO1	BT2
e	Find the volume of the region bounded above by the paraboloid $z = x^2 + y^2$ and below by the triangle enclosed by the lines $y = x$, $x = 0$ & $x + y = 2$ in the xy -plane.	06	CO1	BT2
f	Find the inverse Laplace transform of following functions: i) $F(s) = \frac{1}{s^2 - 5s + 6}$ ii) $F(s) = \frac{6s+7}{s^2 + 2s + 5}$	06	CO4	BT2
Q.4 Attempt all				
a	Find the extreme values of $f(x, y) = x^2 - y^2$ on the circle C of radius 1 centred at the origin in the xy -plane.	06	CO1	BT3
b	Compute divergence and curl of the vector field $F(x, y, z) = 2xye^z \mathbf{i} + e^z x^2 \mathbf{j} + (x^2 ye^z + z^2) \mathbf{k}$.	06	CO1	BT1
c	Find the work done by the force field $F(x, y) = e^{-y} \mathbf{i} - xe^{-y} \mathbf{j}$ in moving an object along the path $c(t) = (2 - 2t, t)$, $0 \leq t \leq 1$.	06	CO1	BT3
d	Find the Fourier cosine series to represent, $f(x) = x^2 + 2$, $-\pi \leq x \leq \pi$ & $f(x + 2\pi) = f(x)$.	06	CO2	BT2
e	Find the Laplace transform of following functions: i) $f(t) = \sin(2t) \cdot \cos(3t)$ ii) $f(t) = t^3 e^{-2t}$ iii) $f(t) = e^{-t} \sin^2 t$.	06	CO4	BT2
Q.3 Attempt all				
a	Find the Fourier series of the periodic function $f(x)$ defined as $f(x) = \begin{cases} 2, & -\pi \leq x < 0 \\ x, & 0 \leq x < \pi \end{cases}$ & $f(x + 2\pi) = f(x)$.	08	CO2	BT2
b	Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{-x}{\sqrt{x^2 + y^2}}$ does not exist. OR Show that the vector field $V(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ cannot be the curl of some vector field F .	02	CO1	BT2

[BT1: Remembering, BT2: Understanding, BT3: Applying, BT4: Analysing, BT5: Evaluating, BT6: Creating]

CO1. Apply the knowledge of multivariable calculus to solve problems related to data Science, optimization, electrical, mechanical, chemical process.

CO2. Apply the concept of Fourier series and their application in system communication and digital signal processing, vibrations.

CO3. Examine the concept of partial differential equations to describe the heat and wave propagation phenomenon arising in fluid mechanics, electromagnetic theory, mechanical, and other engineering problems.

CO4. Use the knowledge of Laplace Transforms of customary functions arising in network analysis and control systems and solve ODE's and PDEs optimization electrical, mechanical, chemical process.



**Shri Guru Gobind Singhji Institute of Engineering and Technology,
Nanded**
Department of BSH&M (Mathematics)
Summer term Examination,
Academic Year: 2024-25 (Semester-I)

Class: FY B.Tech.	Subject: Mathematics-I (Calculus and linear algebra)	Subject Code: BSC103
Date: 18.06.2025	Time: 10:00 to 13:00	Max. Marks: 100

Note: i) calculators are not allowed. ii) This paper contains total two pages

Q.1	Attempt any five	Marks	CO	BT
a	Verify that the following function satisfies the three hypotheses of Rolle's Theorem on the given interval. Find all numbers "c" such that f satisfy the conclusion of Rolle's Theorem. $f(x) = \sqrt{x} - \frac{1}{3}x$, $[0,9]$.	06	CO1	BT2
b	Find the absolute maximum and absolute minimum values of the function on the given interval $f(x) = 3x^2 - 12x + 5$, $[0,3]$.	06	CO1	BT3
c	Find: (a) $\lim_{x \rightarrow \infty} \frac{e^x}{x^2}$ (b) $\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$	06	CO1	BT2
d	Find Taylor's series of $f(x) = (1+x)^{-2}$ about $a = 0$ (ie find Maclaurin series).	06	CO1	BT2
e	i) If $f(1) = 10$ and $f'(x) \geq 2$ for $1 \leq x \leq 4$, how small can $f(4)$ be possible? ii) Use the mean value theorem to prove the inequality $ \cos(a) - \cos(b) \leq a - b , \forall a, b \in \mathbb{R}$.	06	CO1	BT3
f	Solve the following differential equation $\frac{1}{y^2} \frac{dy}{dx} - \frac{1}{y} = 2xe^{-x}$.	06	CO3	BT2
Q.2	Attempt any five			
a	Evaluate the upper and lower sums for $f(x) = 1 + x^2$, $-1 \leq x \leq 1$, with $n = 4$.	06	CO2	BT1
b	i) Use Fundamental theorem of calculus to find $f'(x)$ for the function $f(x) = \int_1^x \frac{1}{t^3+1} dt$. ii) Check the following differential equation is exact or not? $(2xy + e^y)dx + (x^2 + xe^y)dy = 0$	06	CO3	BT2
c	Find rank of the matrix $A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 4 & 1 & 2 & 1 \\ 3 & -1 & 1 & 2 \\ 1 & 2 & 0 & 1 \end{bmatrix}$.	06	CO3	BT2

d	Find all eigenvalues of the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 2 & 2 \\ 1 & 1 & 3 \end{bmatrix}$.	06	CO3	BT2
e	Sketch the region enclosed by the following parabolic curves & find its area: $y = x^2$, $y = -x^2 + 2$.	06	CO2	BT3
f	Find the volume of the solid obtained by rotating the region bounded by the curves $y = x^2$, $y = 0$, $x = 0$ & $x = 1$ about the x -axis.	06	CO2	BT3
Q.3 Attempt all				
a	If $f(x) = 2x^3 + 3x^2 - 36x$ then, (i) Find the interval on which f is increasing or decreasing. (ii) Find the local maximum and local minimum values of f . (iii) Find the interval of concavity and the inflection points.	10	CO1	BT3
b	For what values of λ and μ the system of linear equations: $x + y + z = 6$ $x + 2y + 5z = 10$ $2x + 3y + \lambda z = \mu$ has (i) a unique solution (ii) no solution (iii) infinite solutions (iv) Find the solution for $\lambda = 2$ and $\mu = 8$.	10	CO4	BT3
c	Find all eigenvectors associated to eigenvalues $\lambda_1 = 1$ & $\lambda_2 = 7$ of the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{bmatrix}$. (Do not compute eigenvalues, take eigenvalues $\lambda_1 = 1$ & $\lambda_2 = 7$ for matrix A)	10	CO4	BT2
d	i) Solve the following differential equation by variation of parameters $\frac{d^2y}{dx^2} - y = \frac{2}{1+e^x}$. [you can directly use $\int \frac{1}{e^x(1+e^x)} dx = -e^{-x} + \log(e^{-x} + 1)$]	08	CO3	BT2
	ii) Find the order and degree of the following differential equation $\frac{d^2y}{dx^2} + \frac{dy}{dx} + 8y = \tan x$.	02		

[BT1: Remembering, BT2: Understanding, BT3: Applying, BT4: Analysing, BT5: Evaluating, BT6: Creating]

CO1. Understand the concepts of derivatives and apply the knowledge of differentiation to solve problems related to polar curves, radius of curvature and related engineering problems.

CO2. Learn the definite and improper integrals and apply the notion to engineering problems.

CO3. Learn the notion of ordinary differential equations and use them in electric circuit problem, communication, optimization, etc.

CO4. Make use of matrix theory for solving system of linear equations and compute eigen values and eigenvectors and solve the problems related to his engineering field in data science, electrical, mechanical, chemical process, optimization, etc.